Improving the Efficiency of Subgroup Analysis in Clinical Trials with Adaptive TMLE

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Abstract

1 Introduction

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Estimating treatment effects within specific patient subgroups in clinical trials is crucial for understanding potential treatment effect heterogeneity. However, clinical trials often lack sufficient representation of certain subpopulations, limiting the statistical power of subgroup-specific analyses. Real-world data (RWD) sources offer a potential avenue to augment trial subpopulations and enhance inference for subgroup effects. However, integrating RWD with randomized controlled trial (RCT) data introduces substantial statistical challenges, particularly in controlling bias arising from incompatibilities between data sources. These challenges include distributional shifts and structural differences in how data are generated and recorded, such as variations in patient populations and eligibility criteria, treatment implementation and adherence, endpoint definitions, confounding mechanisms, and coding systems. Therefore, it is of interest to explore methods for improving subgroup analyses using trial data alone, without reliance on external RWD.

When restricting attention to RCT data alone, a natural approach to subgroup analysis is to consider whether information from patients outside the subgroup, yet still enrolled in the trial, can be leveraged to improve estimation of the subgroup-specific treatment effect. In this work, we propose to adapt data integration methods, originally developed for combining RCT data with RWD, to this setting. Specifically, we conceptualize the subgroup as the target population and patients outside the subgroup as an external source, and apply data integration estimators to improve estimation of the treatment effect within the subgroup. This way, we draw on the internal infrastructure of the RCT, where data quality, randomization, and measurement protocols are already harmonized, while correcting for subgroup differences. By recasting the rest of the trial population as "external," we sidestep many of the comparability issues that arise when integrating RCT data with external RWD, while possibly still gaining efficiency in finite sample through statistical borrowing.

Given sample size limitations, efficient estimators are tantamount to subgroup effect estimation within a trial population. Targeted Maximum Likelihood Estimation (TMLE) is an asymptotically efficient estimator with several key desirable features in this setting. The targeted updating and double robustness properties ensure bias reduction and consistent estimation. TMLE also allows for integration of flexible machine

learning algorithms into the nuisance parameter estimation. In the data integration setting specifically, Adaptive Targeted Maximum Likelihood Estimation (A-TMLE) decomposes the target estimand into (i) a pooled estimand, which combines information from the target and external sources, and (ii) a bias estimand, which adjusts for bias induced by incorporating external data.(van der Laan et al., 2023, 2024b) In the context of subgroup analysis, the pooled estimand is naturally defined over the full RCT population, while the bias estimand captures the difference between the subgroup-specific target estimand and the pooled estimand. We note, however, that this framework is not restricted to A-TMLE, and in principle, any data integration estimator developed for RCT-RWD settings could be adapted to this subgroup analysis setting. Our motivation for using A-TMLE in this work includes the following:

1. In a well-designed and properly executed RCT, treatment assignment is randomized both overall and within subgroups. Under this condition, theoretical results for A-TMLE suggest that the estimator is asymptotically unbiased, even under misspecification of nuisance parameters (see Section xxx for further discussion).

2. The magnitude of finite-sample efficiency gains from A-TMLE generally depends on the complexity of the working model for the conditional effect of subgroup membership on the outcome. When the true functional form is relatively simple—for example, constant or well-approximated by a parsimonious function of baseline covariates and treatment—substantial efficiency gains may be achieved. Thus, even when treatment effect heterogeneity is present, if it can be captured by a relatively simple structure, A-TMLE may still offer meaningful improvements in efficiency in finite sample.

In both simulation studies and this real-world application, A-TMLE improves efficiency compared to conventional subgroup-specific estimators. These results highlight A-TMLE as a practical and theoretically grounded tool for enhancing subgroup-specific evidence generation within the existing infrastructure of a clinical trial.

We illustrate the utility and practicality of this strategy using the Liraglutide Effect and Action in Diabetes (LEADER) trial, a phase 3B international, randomized, double-blind, placebo-controlled cardiovascular safety trial sponsored by Novo Nordisk. The trial enrolled patients with type 2 diabetes mellitus (T2DM) and followed them for up to five years, measuring a composite primary endpoint of first evidence of major adverse cardiac event (MACE)Marso et al. (2013). While the trial enrolled 9,340 patients and was adequately powered for its primary safety endpoint, subgroup analyses were generally underpowered (as is common in such settings). No significant subgroup effects by relevant demographic characteristics, such as race, were found in the main trial analysis. While there is no expected biological effect of race, a recent exploratory meta-analysis of relevant glucagon-like peptide 1 receptor agonist (GLP1) cardiovascular trials, including LEADER, found a stronger effect among Asian race than white race. While pooling trials can increase power, it comes with expected drawbacks of treatment and population heterogeneity, estimation challenges, bias...(etc).

To address the question of differential race subgroup effects while maintaining the integrity of the original RCT design, we apply A-TMLE to estimate subgroup-specific treatment effects within the LEADER trial, drawing strength from the remainder of the trial population while adjusting for subgroup-specific differences.

2 Problem setup

We begin by introducing notation for the observed data. We then define the causal estimand of interest, namely the risk difference of MACE between the treatment and control arm at a user-specified time τ among the Asian population. Next, we present a set of identification assumptions under which this causal estimand can be expressed as a functional of the observed data distribution P_0, thereby yielding a statistical estimand. Finally, we derive the canonical gradient of this statistical estimand, which serves as the basis for constructing the A-TMLE estimator.

2.1 Observed data

Let T denote the time to first occurrence of a MACE, C denote the censoring time, and let $\Delta \in \{0, 1\}$ be the event indicator, where $\Delta = 1$ indicates that the event is observed and $\Delta = 0$ indicates right-censoring. The observed data for each individual can thus be written as $O = (S, W, A, \tilde{T} = \min(T, C), \Delta)$, where $S \in \mathcal{S} = \{0, 1\}$ indicates subgroup membership, $W \in \mathcal{W} \subset \mathbb{R}^d$ is a vector of baseline covariates, and $A \in \mathcal{A} = \{0, 1\}$ is a binary treatment. In the LEADER trial, participants were closely monitored with minimal loss to follow-up (Marso et al., 2016). Therefore, we make the simplifying assumption of no loss to follow-up. Our outcome of interest is a binary indicator of MACE occurrence by a fixed time horizon τ, namely, $Y = I(T \leq \tau)$, and we work with the simplified observed data structure $O = (S, W, A, Y)$.

Our observed data consists of n copies of O: $O_1, \ldots, O_n$ drawn i.i.d. from a probability distribution P_0 that lies in a statistical model $\mathcal{M}$. For our statistical model, we have knowledge that the treatment in the trial is randomized, that is, $P(A = 1 \mid S = 1, W) = 1/2$. We assume the following Structural Causal Model (SCM):

$$
\begin{aligned}
S &= f_S(U_S), \\
W &= f_W(S, U_W), \\
A &= f_A(U_A), \\
Y &= f_Y(W, S, A, U_Y),
\end{aligned}
$$

where f_S, f_W, f_Y are unrestricted unknown functions and (U_S, U_W, U_A, U_Y) is a vector of exogenous errors. The function f_A is known and corresponds to a randomized assignment mechanism, that is, e.g., $A = I(U_A \leq 1/2)$ with $U_A \sim \text{Uniform}(0, 1)$, implying that A is independent of (W, S). This SCM encodes knowledge that subgroup membership S may influence baseline covariates W and outcome Y; and baseline covariates W and treatment A may causally affect the outcome Y. Potential outcomes are defined via interventions on the treatment node A. Specifically, for $a \in \mathcal{A}$, we define $Y_1 = f_Y(S, W, A = 1, U_Y)$ and $Y_0 = f_Y(S, W, A = 0, U_Y)$, the outcomes that would have been observed for a randomly sampled individual had they, possibly contrary to fact, been assigned to treatment or control, respectively.

Remark 1. *We note that the notation used here mirrors van der Laan et al. (2024b), apart from S denoting subgroup membership here instead of RCT membership in the original article.*

2.2 Causal estimand and its identification

Our target causal estimand $\Psi : \mathcal{M} \to \mathbb{R}$ is the subgroup-specific risk difference of MACE between the treatment and control arm at time τ, formally defined as

$$\Psi(P) = E_{P_{W|S=1}}[E_P(Y_1 - Y_0 \mid S = 1, W)],$$

where the outer expectation marginalizes over the covariate distribution among individuals in the subgroup of interest ($S = 1$), that is, the Asian population.

Remark 2. *In the original A-TMLE framework for data integration (van der Laan et al., 2024b), the target causal estimand is defined as*

$$E_{P_W}[E_P(Y_1 - Y_0 \mid S = 1)],$$

which differs from our definition in that the outer expectation is taken over the marginal covariate distribution P_W, corresponding to the pooled population (that is, the RCT enrolled population in the current context). In contrast, our goal here is to characterize the treatment effect within the subgroup itself. Accordingly, we define the target estimand with respect to $P_{W|S=1}$, which is more appropriate for subgroup-specific inference. Notably, the estimand Ψ introduced here was also mentioned in van der Laan et al. (2024b), and it is denoted by Ψ_2.

The following identification assumptions are needed to nonparametrically identify the causal estimand:

A1 Consistency within the subgroup. Y_a when $A = a$ for $a \in \mathcal{A}$ given $S = 1$;

A2 No unmeasured confounding in the subgroup. $A \perp (Y_0, Y_1) \mid W, S = 1$;

A3 Positivity of treatment assignment in the subgroup. For all $w \in \mathcal{W}$, $0 < P(A = a \mid S = 1, W = w) < 1$ for $a \in \mathcal{A}$.

In our setting, we view the subgroup of interest as the population in which we intend to draw inference on, and treat observations from the remainder of the RCT population as auxiliary data ($S = 0$) to support estimation. Therefore, due to randomization of the treatment assignment in the RCT, all three assumptions are satisfied by design. Under these assumptions, the causal estimand Ψ is equal to the statistical estimand:

$$\Psi(P) = E_{P_{W|S=1}}[E_P(Y \mid S = 1, W, A = 1) - E_P(Y \mid S = 1, W, A = 0)].$$

2.3 Decomposition of the estimand and its corresponding projection estimand

In the A-TMLE framework, one first decompose the target estimand into a pooled estimand and a bias estimand (van der Laan et al., 2024b). Here, we follow the same decomposition strategy. For any probability distribution $P \in \mathcal{M}$, define

$$Q_P(s, w, a) := E_P(Y \mid S = s, W = w, A = a), \qquad \bar{Q}_P(w, a) := E_P(Y \mid W = w, A = a),$$

and let

$$\Pi_P(s \mid w, a) := P(S = s \mid W = w, A = a), \qquad \tau_{S,P}(w, a) := Q_P(1, w, a) - Q_P(0, w, a).$$

We first define the pooled estimand $\tilde{\Psi}(P) := E_{P_{W|S=1}}[\bar{Q}_P(W,1) - \bar{Q}_P(W,0)]$. This estimand averages over the covariate distribution of the target subgroup, but uses the outcome regression learned from the pooled sample. This estimand is a more efficient target from an estimation perspective because the inner outcome regression allows us to leverage both $S = 1$ and $S = 0$ (i.e., the entire RCT data) to estimate. However, for example, when the Asian population has different baseline risks due to having a different distribution of baseline covariates than the rest of the population or treatment effect heterogeneity, then the estimand $\tilde{\Psi}$ may not necessarily equal to our target of interest, Ψ. To capture this difference, we define the bias estimand as $\Psi^{\#}(P) = \tilde{\Psi}(P) - \Psi(P)$, which captures precisely the difference between the pooled estimand and our target estimand. It has been shown in (van der Laan et al., 2024b) (replacing the outer expectation P_W with $P_{W|S=1}$ in the proof of Lemma 2) that the bias estimand $\Psi^{\#}$ has an analytic form:

$$\Psi^{\#}(P) := E_{P_{W|S=1}}[\Pi_P(0 \mid W,0)\tau_{S,P}(W,0) - \Pi_P(0 \mid W,1)\tau_{S,P}(W,1)],$$

a combination of the bias function $\tau_{S,P}$ weighted by the subgroup enrollment probability Π_P. Now, in order to obtain an estimator for Ψ, one may obtain estimators for $\tilde{\Psi}$ and $\Psi^{\#}$, respectively, and take the difference.

A key feature of A-TMLE is that instead of directly estimating nonparametrically the estimands $\tilde{\Psi}$ and $\Psi^{\#}$, it estimates the so-called projection estimands. We now define the projection estimands. Let $\mathcal{T}_{A,w} = \{\tau_{A,\beta} : \beta \in \mathcal{B}_A\}$ be a working model for the pooled conditional average treatment effect (CATE) $\tau_{A,P}(w) := \bar{Q}_P(w,1) - \bar{Q}_P(w,0)$, and let the associated semiparametric regression working model be $\bar{\mathcal{Q}}_w = \big\{\theta(w) + a\,\tau_{A,\beta}(w) : \theta, \beta\big\}$, where $\theta(\cdot)$ is some function of w. Here, the subscript w denotes "working model"; throughout the article, any appearance of w in the subscript indicates that the corresponding object is associated with a working model. Let $\bar{g}_P(a \mid W) := P(A = a \mid W)$. Define $\tau_{A,\beta_A(P)}$ as the weighted squared-error projection of $\tau_{A,P}$ onto $\mathcal{T}_{A,w}$:

$$\beta_A(P) \in \arg\min_{\beta} E_P\Big[\bar{g}_P(1 \mid W)\{1 - \bar{g}_P(1 \mid W)\}\big\{\tau_{A,P}(W) - \tau_{A,\beta}(W)\big\}^2\Big].$$

Equivalently, this is the projection induced by the squared-error projection of $\bar{Q}_P$ onto $\bar{\mathcal{Q}}_w$ (Crump et al., 2006; Li et al., 2019; D'Amour et al., 2021; Morzywolek et al., 2023; van der Laan et al., 2023). The corresponding projection parameter for $\tilde{\Psi}$ is then

$$\tilde{\Psi}_{\mathcal{M}_{A,w}}(P) := E_{P_{W|S=1}}\big[\tau_{A,\beta_A(P)}(W)\big].$$

Similarly, let $\mathcal{T}_{S,w} = \{\tau_{S,\beta} : \beta \in \mathcal{B}_S\}$ be a working model for the conditional average subgroup effect $\tau_{S,P}(W,A)$, with associated semiparametric regression working model $\mathcal{Q}_w = \big\{\theta(W,A) + S\,\tau_{S,\beta}(W,A) : \theta, \beta\big\}$, where $\theta(\cdot,\cdot)$ is now a function of (w,a). Define $\tau_{S,\beta_S(P)}$ as the weighted squared-error projection of $\tau_{S,P}$ onto $\mathcal{T}_{S,w}$:

$$\beta_S(P) \in \arg\min_{\beta} E_P\Big[\Pi_P(1 \mid W,A)\{1 - \Pi_P(1 \mid W,A)\}\big\{\tau_{S,P}(W,A) - \tau_{S,\beta}(W,A)\big\}^2\Big].$$

Equivalently, this is induced by the squared-error projection of Q_P onto $\mathcal{Q}_w$. The corresponding projection parameter for $\Psi^{\#}$ is then

$$\Psi^{\#}_{\mathcal{M}_{S,w}}(P) := E_{P_{W|S=1}}\Big[\Pi_P(0 \mid W,0)\tau_{S,\beta_S(P)}(W,0) - \Pi_P(0 \mid W,1)\tau_{S,\beta_S(P)}(W,1)\Big].$$

We therefore obtain the projected version of our target parameter:

$$\Psi_{\mathcal{M}_w}(P) := \tilde{\Psi}_{\mathcal{M}_{A,w}}(P) - \Psi^{\#}_{\mathcal{M}_{S,w}}(P),$$

where $\mathcal{M}_{A,w} := \{P \in \mathcal{M} : \tau_{A,P} \in \mathcal{T}_{A,w}\}$, $\mathcal{M}_{S,w} := \{P \in \mathcal{M} : \tau_{S,P} \in \mathcal{T}_{s,w}\}$, and $\mathcal{M}_w := \mathcal{M}_{A,w} \cap \mathcal{M}_{S,w}$.

2.4 Canonical gradient of the projection estimand $\Psi_{\mathcal{M}_w}$ at $P \in \mathcal{M}_w$

We consider the canonical gradient of $\tilde{\Psi}_{\mathcal{M}_{A,w}}$ (Lemma 2.1) and $\Psi^{\#}_{\mathcal{M}_{S,w}}$ (Lemma 2.2), respectively.

Lemma 2.1. *We define*

$$I_P = E_P\Big[\bar{g}_P(1-\bar{g}_P)(1 \mid W)\phi_A\phi_A^\top(W)\Big] \quad \textit{and}$$
$$\bar{Q}_{w,P}(W,A) = E_P(Y \mid W) + (A - \bar{g}_P(1 \mid W))\sum_j \beta_P(j)\phi_{A,j}(W).$$

The canonical gradient of $\tilde{\Psi}_{\mathcal{M}_{A,w}}$ at P is given by

$$D_{\tilde{\Psi},\mathcal{M}_{A,w},P} = \frac{S}{P(S=1)}\Big[\tau_{A,\beta_{A,P}}(W) - \tilde{\Psi}_{\mathcal{M}_{A,w}}(P)\Big] + \frac{1}{P(S=1)}\Big\{E_P[S\phi_A(W)]\Big\}^\top D_{A,\beta_A,P},$$

where

$$D_{A,\beta_A,P} = I_P^{-1}(A - \bar{g}_P(1 \mid W))\phi_A(W)(Y - \bar{Q}_{w,P}(W,A)).$$

Lemma 2.2. *Let $\bar{\Pi}_P(W) := P(S=1 \mid W)$. We define*

$$\tilde{I}_P = E_P\Pi_P(1-\Pi_P)(1 \mid W,A)\phi_S\phi_S^\top(W,A) \quad \textit{and}$$
$$Q_{w,P}(S,W,A) = E_P(Y \mid W,A) + (S - \Pi_P(1 \mid W,A))\sum_j \beta_{S,P}(j)\phi_{S,j}(W,A).$$

The canonical gradient of the bias projection parameter $\Psi^{\#}_{\mathcal{M}_{S,w}}$ at P is given by

$$D_{\Psi^\#,\mathcal{M}_{S,w},P} = D_{\Psi^\#,\mathcal{M}_{S,w},P_W,P} + D_{\Psi^\#,\mathcal{M}_{S,w},\Pi,P} + D_{\Psi^\#,\mathcal{M}_{S,w},\beta_S,P},$$

where

$$D_{\Psi_2^\#,\mathcal{M}_{S,w},P_W,P} = \frac{S}{P(S=1)}\Big[\Pi_P(0 \mid W,0)\tau_{S,\beta_{S,P}}(W,0) - \Pi_P(0 \mid W,1)\tau_{S,\beta_{S,P}}(W,1)\Big] - \Psi^{\#}_{\mathcal{M}_{S,w}}(P),$$
$$D_{\Psi_2^\#,\mathcal{M}_{S,w},\Pi,P} = \frac{\bar{\Pi}_P(W)}{P(S=1)}\left[\frac{A}{\bar{g}_P(1 \mid W)}\tau_{S,\beta_{S,P}}(W,1) - \frac{1-A}{\bar{g}_P(0 \mid W)}\tau_{S,\beta_{S,P}}(W,0)\right](S - \Pi_P(1 \mid W,A)),$$
$$D_{\Psi_2^\#,\mathcal{M}_{S,w},\beta,P} = \frac{1}{P(S=1)}\Big[E_P[S\cdot\Pi_P(0 \mid W,0)\phi_S(W,0)] - E_P[S\cdot\Pi_P(0 \mid W,1)\phi_S(W,1)]\Big]^\top D_{S,\beta_S,P}.$$

where

$$D_{S,\beta_S,P} = \tilde{I}_P^{-1}(S - \Pi_P(1 \mid W,A))\phi_S(W,A)(Y - Q_{w,P}(S,W,A)).$$

Remark 3. *The canonical gradient of the nonparametrically defined target estimand Ψ at $P \in \mathcal{M}$ is given*

by

$$D_{\Psi,P} = \frac{S}{P(S=1)}\Big[Q_P(1,W,1) - Q_P(1,W,0) - \Psi(P)\Big] + \frac{S}{P(S=1)} \cdot \frac{2A-1}{g_P(A \mid 1, W)}\Big[Y - Q_P(S,W,A)\Big].$$

A standard TMLE for Ψ can be constructed based on this canonical gradient. In implementing such a standard TMLE, there are multiple strategies for estimating the outcome regression $Q_P(1,W,a)$. One approach is to restrict estimation to the subgroup data ($S=1$) by fitting the model using only this subset. Alternatively, one may fit the outcome regression using the full RCT sample and evaluate predictions at $S=1$. Furthermore, one may define a discrete Super Learner (van der Laan et al., 2007) to data-adaptively select between these two strategies. We defer a detailed investigation of this standard TMLE to future work.

HERE WE NEED SOME DISCUSSION ON THE THEORETICAL DICUSSION ON THE DR STRUCTURE OF ATMLE FROM OUR 2ND PAPER. SHOW THAT IN THIS CONTEX, WE CAN GET ASYMP. UNBIASENESS WRT NP DEFINED ESTIMAND WITH ARBITRARY MISSPECIFICATION OF THE WORKING MODELS.

3 Constructing an adaptive TMLE and practical considerations

To make this work self-contained, we now provide a brief review of the key elements of constructing an A-TMLE.

3.1 Strategies for estimating nuisance functions and working models

The construction of an A-TMLE estimator requires estimators for the working models. We use the loss function

$$\tilde{L}_{\theta_P,\bar{g}_P}(O)(\tau) = \Big[Y - \theta_P(W) - (A - \bar{g}_P(1 \mid W))\tau\Big]^2$$

to learn the pooled conditional average treatment effect τ_{A,P_0}, where the loss function is indexed by nuisance functions $\theta_P(W) = E_P(Y \mid W)$ and $\bar{g}_P = P(A=1 \mid W)$.

Remark 4. *The loss function $\tilde{L}$ is double robust with respect to misspecification of either θ or $\bar{g}$ in that $\tau_{A,\beta_{A,P}} = \arg\min_{\tau \in \mathcal{T}_{A,w}} E_P \tilde{L}_{\theta,\bar{g}}(O)(\tau)$ if either $\theta = \theta_P$ or $\bar{g} = \bar{g}_P$.*

Similarly, we use the loss function

$$L^{\#}_{\bar{Q}_P,\Pi_P}(O)(\tau) = \Big[Y - \bar{Q}_P(W,A) - (S - \Pi_P(1 \mid W,A))\tau\Big]^2$$

to learn the conditional average subgroup effect τ_{S,P_0}, where the loss function is indexed by nuisance functions $\bar{Q}_P(W,A) = E_P(Y \mid W,A)$ and $\Pi_P(1 \mid W,A) = P(S=1 \mid W,A)$.

Remark 5. *The loss function $L^{\#}$ is double robust with respect to misspecification of either $\bar{Q}$ or Π in that $\tau_{S,\beta_{S,P}} = \arg\min_{\tau \in \mathcal{T}_{S,w}} E_P L^{\#}_{\bar{Q},\Pi}(O)(\tau)$ if either $\bar{Q} = \bar{Q}_P$ or $\Pi = \Pi_P$.*

We recommend estimating the nuisance functions θ, $\bar{g}$, $\bar{Q}$, and Π using Super Learner, an ensemble approach that allows the incorporation of multiple flexible machine learning algorithms (van der Laan et al.,

2007). For obtaining working models for conditional average effects τ_{A,β_A} and τ_{S,β_S}, we recommend using Highly Adaptive Lasso (HAL) (van der Laan, 2017; Benkeser and van der Laan, 2016).

HAL is a nonparametric regression approach that attempts to represent an unknown target function by a very large dictionary of simple basis functions and then estimates the coefficients by empirical risk minimization with an ℓ_1-type penalty. HAL only assumes that the true function f_0 is *càdlàg* (right continuous with left limits) with a bounded variation (as defined in van der Laan (2017)), which is arguably a plausible assumption for most functions one may encounter in the domains such as public health. It is motivated by a general representation of multivariate càdlàg functions, and HAL acts as an empirical risk minimizer in that function class. Model complexity is controlled through the ℓ_1-norm of the basis coefficients, which corresponds to the *sectional variation norm* of the function, a natural quantity for regularization.

For a function $f_0 \in D^{(0)}([0,1]^d)$, the zero-order HAL basis dictionary consists of indicator basis functions indexed by coordinate subsets and knot points.

More concretely, if $\phi_1, \ldots, \phi_J$ denotes a HAL dictionary of J basis functions (including an intercept term) and $L(\cdot)$ denotes a loss function, HAL estimates a regression function Q_0 by solving

$$Q_n = \arg\min_{Q_\beta} P_n L(Q_\beta) \quad \text{subject to} \quad Q_\beta = \sum_{j=1}^{J} \beta_j \phi_j, \qquad \sum_{j=1}^{J} |\beta_j| \leq C_n,$$

or, equivalently, by a Lasso-penalized empirical risk minimization problem. The parameter C_n controls the complexity of the fit and is typically selected by cross-validation. HAL is attractive because it also provides us with a working model defined by the basis functions with non-zero coefficients in the final fit.

As an example, suppose f_0 is a tri-variate function with arguments X_1, X_2, and X_3. Suppose we do not have knowledge on the smoothness of the function, and $f_0 \in D^{(0)}([0,1]^d)$, where $D_M^{(0)}([0,1]^d)$ is the cadlag function class with no smoothness assumption and with a bound M on the sectional variation norm. Then a zero-order HAL basis dictionary would consist of indicator basis functions of the form one-way basis functions $I(X_1 \geq u)$, $I(X_2 \geq u)$, $I(X_3 \geq u)$ for $u \in [0,1]$; two-way basis functions $I(X_1 \geq u_1, X_2 \geq u_2)$, $I(X_1 \geq u_1, X_3 \geq u_2)$, $I(X_2 \geq u_1, X_3 \geq u_3)$ for $(u_1, u_2) \in [0,1]^2$; and three-way basis functions $I(X_1 \geq u_1, X_2 \geq u_2, X_3 \geq u_3)$ for $(u_1, u_2, u_3) \in [0,1]^3$. In Benkeser and van der Laan (2016), the authors considered to use a set of noninformative knot points implied by the observed data. However, in practice, it is often recommended to limit the maximum degrees of interactions and number of knot points per subspace for computational feasibility consideration. In Subsection 3.3, we discuss a novel approach to perform such screening that is theoretically motivated. Therefore, one could first generate the HAL design matrix with each column a HAL basis function evaluated at the observed data and then run off-the-shelf Lasso optimization software such as the `R` package `glmnet`.

3.2 TMLE-style targeting of the working model coefficients of $\tilde{\Psi}$ and $\Psi^{\#}$

In van der Laan et al. (2024b), the authors propose to refit the working-model coefficients unpenalized so that the empirical mean of the corresponding components of the EIC is solved to zero. Here, we propose an alternative, TMLE-style approach for targeting these coefficients.

Let's first consider the parameter $\tilde{\Psi}$. The following lemma

Lemma 3.1. *Define $C_n = I_n^{-1} E_n \phi_A(W)$ as the clever covariate. Let*

$$H_n = (A - g_n(1 \mid W))\phi(W) C_n, \qquad R_n = Y - \theta_n(W) - (A - g_n(1 \mid W))\tau_{A,\beta_n}(W).$$

Given a one-dimensional linear fluctuation submodel $\epsilon \to \beta_{n,\epsilon} = \beta_n + \epsilon C_n$ for β through its initial β_n in the direction C_n. The closed-form solution $\epsilon^\star$ of ϵ that solves the score equation $0 = P_n H_n(R_n - \epsilon H_n)$ is given by

$$\epsilon^\star = \frac{P_n(H_n R_n)}{P_n(H_n^2)}.$$

The proof is provided in Appendix B.1. Lemma 3.1 shows that it is not necessary to solve all scores equations in the working model using the unpenalized fit. Since our goal is to solve the empirical mean of the EIC, we only need to solve that particular one. This preserves all the signals to solve the score equation that really matters from a de-biasing perspective.

Now, similarly, we can derive a similar lemma for β_S for the working model for $\Psi^\#$.

Lemma 3.2. *Define a one-dimensional linear fluctuation submodel $\gamma \to \beta_{S,n,\gamma} = \beta_{S,n} + \gamma C_{S,n}$ for β_S in the direction $C_{S,n}$. Let*

$$\begin{aligned} H_{S,n} &= (S - \Pi_n(1 \mid W, A))\phi_S(W, A) C_{S,n} \\ R_{S,n} &= Y - \theta_n(W, A) - (S - \Pi_n(1 \mid W, A))\tau_{S,\beta_n}(W, A). \end{aligned}$$

The closed-form solution $\gamma^\star$ of γ that solves the estimating equation $0 = P_n H_{S,n}(R_{S,n} - \gamma H_{S,n})$ is given by

$$\gamma^\star = \frac{P_n(H_{S,n} R_{S,n})}{P_n(H_{S,n}^2)}.$$

The proof is provided in Appendix B.2.

3.3 Using group lasso to screen for HAL subspaces

For zero-order HAL, assuming all the covariates are continuous, the dimension of the HAL design matrix could go as large as $n \times (2^d - 1)$, which is an astronomically large and incomputable. Therefore, it is *never* recommended to run the full-blown HAL in practice. There have been some recent work addressing the computational aspect of HAL, including Schuler et al. (2022); Wang et al. (2026); Meixide et al. (2026). Here, we introduce a novel approach that is both practically and theoretically appealing to screen the number of HAL basis functions down to a computationally feasible regime via a group-sparse penalty.

The A-TMLE framework suggests viewing $D^{(k)}(\mathcal{R}_n)$ as a data-adaptive working model and requiring it to stabilize relative to a fixed oracle model $D^{(k)}(\mathcal{R}_0)$ (van der Laan et al., 2023). A convenient regime is the special case $P\big(D^{(k)}(\mathcal{R}_n) \subseteq D^{(k)}(\mathcal{R}_0)\big) \to 1$, together with the requirement that the sieve within $D^{(k)}(\mathcal{R}_n)$ approximates $D^{(k)}(\mathcal{R}_0)$ from below at the required rate. This is the analogue of the ADML conditions that the learned working influence function converges to the oracle influence function and that the model-approximation error is negligible, i.e., the analogues of Conditions C2 and C4 in van der Laan et al. (2023). In the case when the oracle model $\mathcal{R}_0$ is unknown and represents the support of the true additive or interaction structure of Q_0. In this setting, the goal is not necessarily exact recovery of the smallest support, but rather asymptotic stabilization to some fixed oracle set $\mathcal{R}_0$ such that $Q_0 \in D^{(k)}(\mathcal{R}_0)$. The oracle model need not be minimal; it may simply be the limiting selected model. A natural sufficient formulation is therefore that $\mathcal{R}_n$ converges from below to $\mathcal{R}_0$, in the sense that $P(\mathcal{R}_n \subseteq \mathcal{R}_0) \to 1$, and the induced sieve within $D^{(k)}(\mathcal{R}_n)$ becomes sufficiently rich to approximate $D^{(k)}(\mathcal{R}_0)$ at the required rate. Under this regime, the final estimator lies in the oracle subspace with high probability for large n.

HAL dictionary is naturally organized by *subspace*. Each basis function depends on a particular subset of coordinates $s \subset \{1, \ldots, d\}$ and a knot point. Therefore, the collection of all basis functions sharing the same coordinate set s can be viewed as representing one additive or interaction component of the target function. Now, this motivates a practical way to construct such a sequence $\mathcal{R}_n$ using a group-sparsity penalty as a first-stage screen for subspaces. Consider the zero-order HAL case with $k = 0$. Suppose we restrict attention to variable interactions of degree at most m, giving the collection of coordinate indices

$$\mathcal{S}_m = \{s \subset \{1, \ldots, d\} : 0 < |s| \leq m\}.$$

For each $s \in \mathcal{S}_m$, let β_s denote the block of coefficients corresponding to all zero-order HAL basis functions depending on the same coordinate set s but different knot points. Given a loss function $L(Q)(O)$, one may define the group lasso estimator (Yuan and Lin, 2006) given by

$$\hat{\beta} = \arg\min_{\beta} P_n L(Q_\beta)(O) + \lambda \sum_{s \in \mathcal{S}_m} w_s \|\beta_s\|_2,$$

with weights w_s chosen, for example, as $w_s = \sqrt{p_s}$, where p_s is the number of basis functions in group s. The role of this penalty is to induce sparsity at the level of subspaces rather than individual basis functions. That is, for many interaction sets s, the entire coefficient block β_s is set to zero. In this sense, the group lasso acts as a screening rule for additive components and interactions, producing a relatively small collection of candidate subspaces that appear to matter.

This screening step can then be combined with a second-stage HAL-MLE. Specifically, one may first run the group lasso with a small number of knot point for each subspace, over a range of λ values, to generate a ranked family of candidate supports $\mathcal{R}_{0,\ell}$. Then, for each candidate support, one fits a HAL-MLE restricted to the corresponding subspace $D^{(k)}(\mathcal{R}_{0,\ell})$, using a much denser knot point grid. Cross-validation may then be used over the candidate support index ℓ to select the final working model. Conceptually, this mirrors the setup of cross-validation over a growing sequence of working models described in van der Laan et al. (2023). Specifically, the screening step selects a candidate oracle submodel, and the final within-subspace HAL fit may be undersmoothed, possibly, only relative to that selected submodel, rather than relative to the full function space $D^{(k)}([0,1]^d)$.

We present the pseudo-code for the implementation of A-TMLE in Algorithm 1. NEED TO UPDATE THIS, NEW TARGETING METHOD. ALSO PARAM NOW AVG OVER S=1.

4 Simulation study

The goal of this simulation study is to demonstrate that A-TMLE is able to achieve efficiency gain, measured in MSE, as compared to a standard practice for subgroup analysis applying an estimator on the subgroup specific data, while preserve nominal confidence interval coverage. Considering our LEADER real-data application, we designed our simulation study to closely mimic the data-generating distribution that likely generated the real LEADER trial data. Due to sensitive nature of patient-level data, we are not able to share the patient-level data. Therefore, to make the simulation still reproducible, we generate baseline covariates based on summary statistics and dependence structures of the observed LEADER trial data.

Algorithm 1 Adaptive TMLE

Obtain estimators g_n of g_0 and θ_n of θ_0;
Fit a HAL with basis functions $\{\phi_j(W) : j\}$:

$$\alpha_n = \arg \min_{\alpha, \|\alpha\|_1 \leq C_n} \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - \theta_n(W_i) - (A_i - g_n(1 \mid W_i)) \sum_j \alpha(j)\phi_j(W_i) \right)^2;$$

Define $\mathcal{R}_\alpha$ as the index set of α_n with non-zero entries and compute the OLS in the working model $\{\phi_j(W) : j \in \mathcal{R}_\alpha\}$:

$$\alpha_n^\star = \arg \min_{\alpha} \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - \theta_n(W_i) - (A_i - g_n(1 \mid W_i)) \sum_{j \in \mathcal{R}_\alpha} \alpha(j)\phi_j(W_i) \right)^2;$$

Obtain estimators $\Pi_n^{(0)}$ of Π_0 and $\bar{Q}_n$ of $\bar{Q}_0$;
Fit a HAL with basis functions $\{\phi_j(W, A) : j\}$:

$$\beta_n^{(0)} = \arg \min_{\beta, \|\beta\|_1 \leq C_n^\beta} \frac{1}{n} \left(Y_i - \bar{Q}_n(W_i, A_i) - (S_i - \Pi_n^{(0)}(1 \mid W_i, A_i)) \sum_j \beta(j)\phi_j(W_i, A_i) \right)^2;$$

Define $\mathcal{R}_\beta$ as the index set of β_n with non-zero entries, and define convergence threshold $s_n := \sigma_n^{(0)}/(\sqrt{n} \cdot \log n)$, where $\sigma_n^{(0)}$ is the estimated variance of the EIC of Ψ at $P_n^{(0)} = (g_n, \theta_n, \alpha_n^\star, \Pi_n^{(0)}, \bar{Q}_n, \beta_n^{(0)})$. Set iteration counter $k := 0$;
while $|P_n D_{\mathcal{M}_w, P_n^{(k)}}| > s_n$ **do**
    Target Π by fitting a univariate logistic regression:

$$\operatorname{logit} \Pi_n^{(k+1)} = \operatorname{logit} \Pi_n^{(k)} + \epsilon_n^{(k)} \left(\frac{A}{g_n(1 \mid W)} \tau_{S, \beta_n^{(k)}}(W, 1) - \frac{1 - A}{g_n(0 \mid W)} \tau_{S, \beta_n^{(k)}}(W, 0) \right),$$

    where $\epsilon_n^{(k)}$ is the estimated coefficient of this univariate logistic regression;
    Target β by computing the OLS in the working model $\{\phi_j(W) : j \in \mathcal{R}_\alpha\}$:

$$\beta_n^{(k+1)} = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - \bar{Q}_n(W_i, A_i) - (S_i - \Pi_n(1 \mid W_i, A_i)) \sum_{j \in \mathcal{R}_\beta} \beta(j)\phi_j(W_i, A_i) \right)^2;$$

    Update convergence threshold $s_n := \sigma_n^{(k+1)}/(\sqrt{n} \cdot \log n)$, where $\sigma_n^{(k+1)}$ is the estimated variance of the EIC of Ψ at $P_n^{(k+1)} = (g_n, \theta_n, \alpha_n^\star, \Pi_n^{(k+1)}, \bar{Q}_n, \beta_n^{(k+1)})$;
    Increase iteration counter k by 1;
end while
Define $K := k$, compute the plug-in estimator $\Psi_{\mathcal{M}_w}(P_n^{(K)}) = \tilde{\Psi}_{\mathcal{M}_{A,w}}(\alpha_n^\star) - \Psi^{\#}_{\mathcal{M}_{S,w}}(\Pi_n^{(K)}, \beta_n^{(K)})$, where

$$\tilde{\Psi}_{\mathcal{M}_{A,w}}(\alpha_n^\star) = \frac{1}{n} \sum_{i=1}^{n} \tau_{A, \alpha_n^\star}(W_i), \quad \text{and}$$

$$\Psi^{\#}_{\mathcal{M}_{S,w}}(\Pi_n^{(K)}, \beta_n^{(K)}) = \frac{1}{n} \sum_{i=1}^{n} [\Pi_n^{(K)}(0 \mid W_i, 0)\tau_{S, \beta_n^{(K)}}(W_i, 0) - \Pi_n^{(K)}(0 \mid W_i, 1)\tau_{S, \beta_n^{(K)}}(W_i, 1)]$$

Compute the Wald-type 95% confidence interval given by $\Psi_{\mathcal{M}_w}(P_n^{(K)}) \pm 1.96\sqrt{P_n \left(D_{\mathcal{M}_w, P_n^{(K)}} \right)^2 / n}$.

4.1 Synthetic data generating processes based on LEADER

We generated baseline covariates to resemble the LEADER trial population using a Gaussian copula model with empirically estimated dependence structure and marginal distributions matched to reported summary statistics of patient baseline characteristics of LEADER. Specifically, let $Z = (Z_1, \ldots, Z_d)$ denote a multivariate normal random vector with mean zero and correlation matrix R. We transformed each component to a uniform variable $U_j = \Phi(Z_j)$, where $\Phi(\cdot)$ is the standard normal cumulative distribution function (CDF). These uniforms were then mapped to observed baseline covariates. For continuous baseline covariates, we used a piecewise-linear inverse CDF constructed from the reported minimum, first quartile, median, third quartile, and maximum. Thus, if W_j denotes a continuous covariate and $q_j(p)$ its approximate quantile function, then $W_j = \tilde{q}_j(U_j)$, where $\tilde{q}_j$ is the linear interpolation through the five reported quartile points. This process was done for age, diabetes duration, estimated glomerular filtration rate, body mass index (BMI), HbA1c, low-density lipoprotein (LDL) cholesterol, and high-density lipoprotein (HDL) cholesterol. For binary baseline covariates, we used Bernoulli draws induced by the latent uniforms, i.e., $W_j = I(U_j < p_j)$, where p_j was chosen to match the reported marginal prevalence. This process was done for sex, insulin-naive status, antihypertensive treatment, and prior cardiovascular disease status.

Asian race was generated separately from the Gaussian copula model as a subgroup indicator using a logistic model depending on baseline BMI, so that the simulated Asian subgroup was both approximately the same size as the LEADER trial and leaner on average. Let S denote the Asian race and let z_{BMI} be standardized BMI. We generated $S \sim \text{Bernoulli}\{\text{expit}(\alpha_0 + \alpha_1 z_{\text{BMI}})\}$, where α_0 was chosen so that the marginal prevalence of Asian participants matched the LEADER proportion, that is, 0.076, and $\alpha_1 = \log(0.6) < 0$ made Asian race more common among individuals with lower BMI.

The treatment was randomized independently of baseline covariates, that is, $A \sim \text{Bernoulli}(1/2)$. We designed the following three scenarios for the outcome process:

Scenario 1: Simple proportional-hazard. In this scenario, we generated time to event from a Weibull proportional-hazard (PH) model. Conditional on treatment A and baseline covariates W, the hazard function took the form

$$\lambda_1(t \mid A, W) = \eta \nu t^{\nu-1} \exp(\beta_A A + \beta_W^\top W),$$

where $\nu = 1.5$ is the Weibull shape parameter, η is the scale parameter, $\beta_A = \log(0.87)$ represents a constant treatment hazard ratio of 0.87, and β_W are the coefficients for the selected prognostic baseline covariates. The Weibull scale parameter η is calibrated by Monte Carlo so that, under 1:1 randomization, the overall pooled event risk by 3.8 years matches the LEADER target event proportion, 1302/9340. Event times were simulated by inverse transform sampling:

$$T = \left\{ \frac{-\log(U)}{\eta \exp(\beta_A A + \beta_W^\top W)} \right\}^{1/\nu},$$

where $U \sim \text{Uniform}(0, 1)$. To simplify the design, we assumed no dropout or loss to follow-up. LEADER actually had very high retention and nearly complete endpoint ascertainment. In published follow-up summaries, 96.8% of randomized participants either completed the final visit, died, or experienced a primary outcome event, and vital status was known for 99.7% of participants (Marso et al., 2016). In our simulation study, all participants were administratively censored at a common fixed follow-up time $\tau = 3.8$ years, chosen to reflect the median follow-up in LEADER. Thus, the observed follow-up time and event indicator

were $\tilde{T} = \min(T, \tau)$ and $\Delta = I(T \leq \tau)$. The binary observed outcome used in the simulation was therefore $Y = I(T \leq \tau) = \Delta$, that is, whether the event occurred by 3.8 years.

Scenario 2: Complex proportional-hazard. In this scenario, the hazard still satisfies the PH assumption, but it involves non-linear terms. First, we define the centered/scaled covariates:

$$\text{age}_c = \frac{\text{AGE} - 64}{10}, \quad \text{dur}_c = \frac{\text{DIABDUR} - 11.5}{10}, \quad \text{egfr}_c = \frac{\text{EGFREPB} - 84.3}{20},$$
$$\text{bmi}_c = \frac{\text{BMIBL} - 31.7}{10}, \quad \text{hba1c}_c = \frac{\text{HBA1CBL} - 8.3}{2}, \quad \text{ldl}_c = \text{LDL1BL} - 2.20.$$

Define a non-linear function:

$$h(A, W) = 0.08\text{age}_c^2 + 0.06\text{dur}_c + 0.08\sin(\pi\text{hba1c}_c) + 0.06\text{ldl}_c\text{egfr}_c$$
$$+ 0.08\text{PRCVFL} + 0.06\text{INSNVFL} + 0.04\text{ASIAN}.$$

We then define the log-relative hazard as $f_2(A, W) = \beta_A A + h(W)$, with $\beta_A = \log(0.87)$. So, the hazard is $\lambda_2(t \mid A, W) = \eta\nu t^{\nu-1} \exp\{f_2(A, W)\}$. This is still a PH model because the entire treatment effect enters as a constant additive term $\beta_A A$ in the log-hazard.

Scenario 3: Complex non-proportional-hazard. We now design a scenario in which the hazard is piecewise in time, that is, the PH assumption is violated. First, we define two treatment coefficients $\beta_A^{(1)} = \log(0.95)$ and $\beta_A^{(2)} = \log(0.8)$. Let the change-point $t_0 = 1.5$ years. We define the piecewise hazard:

$$\lambda_3(t \mid A, W) = \begin{cases} \eta\nu t^{\nu-1} \exp\{h(W) + \beta_A^{(1)} A\}, & t < t_0; \\ \eta\nu t^{\nu-1} \exp\{h(W) + \beta_A^{(2)} A\}, & t \geq t_0. \end{cases}$$

So, the treatment effect is weak early and stronger later.

For all three scenarios, the true causal estimand were evaluated by Monte Carlo integration under the known data-generating processes. Specifically, we generate a large sample of baseline covariates W from the same baseline covariate distribution used in the simulation, and for each individual computed the counterfactual event risk under treatment $a \in \mathcal{A}$ at time t as $F_j(t \mid A = a, W) = 1 - \exp\{-\Lambda_j(t \mid A = a, W)\}$, where $\Lambda_j(t \mid A = a, W)$ is the cumulative hazard for the j-th scenario. In the two PH scenario, this reduces to $\Lambda_j(t \mid A = a, W) = \eta t^{\nu} \exp\{f_j(a, W)\}$ for $j \in \{1, 2\}$. In the non-PH scenario, i.e., $j = 3$, $\Lambda_j(t \mid A = a, W)$ was computed using the piecewise cumulative hazard implied by the time-varying treatment effect. The marginal causal risk under treatment level a was then obtained by averaging $F_j(t \mid A = a, W)$ over the Asian patients in the Monte Carlo sample, $R_j(t, a) = E[F_j(t \mid A = a, W) \mid S = 1]$, and the target causal estimand was computed as the risk difference $R_j(t, 1) - R_j(t, 0)$.

4.2 Candidate estimators

Under the three simulation scenarios, we evaluated the performance of A-TMLE against an augmented inverse probability of treatment weighted (A-IPTW) estimator.

Configurations of the A-TMLE. We use a discrete Super Learner (SL) for estimating all nuisance functions using V-fold cross validation scheme with $V = 3$. The SL library we use are the same for both, including main-term generalized linear model (Nelder and Wedderburn, 1972), generalized additive model (Hastie and Tibshirani, 1986; Hastie, 2017), and multivariate adaptive regression splines (Friedman, 1991). For estimating the two CATE functions, i.e., τ_A and τ_S, effect of A on Y and effect of S on Y, we use HAL with the group lasso subspace screening procedure described in Section xxx. For the targeting of β_A and β_S, we use the TMLE-style targeting described in Section xxx.

Configurations of the A-IPTW. We implemented A-IPTW using the R package `AIPW`. A-IPTW uses Super Learner (SL) for estimating nuisance functions, in this case, the outcome regression $E(Y \mid S = 1, W, A)$ and treatment mechanism $P(A = 1 \mid S = 1, W)$. The SL library we use are the same for both, including main-term generalized linear model (`'SL.glm'`), generalized additive model (`'SL.gam'`), and multivariate adaptive regression splines (`'SL.earth'`). The justification for using this SL library is that they run relatively fast given the sample sizes and dimension of the data in the simulation. We use V-fold cross validation, with $V = 3$.

Scenario	n	Estimator	Abs. Bias	SE	MSE	Coverage	Oracle Coverage
Scenario 1	1000	AIPW	0.014055	0.260004	0.067664	0.95	0.95
		A-TMLE	0.001524	0.036659	0.001344	0.92	0.97
	1500	AIPW	0.009254	0.129381	0.016792	0.96	0.93
		A-TMLE	0.001970	0.027381	0.000752	0.93	0.97
	2000	AIPW	0.006794	0.093093	0.008695	0.96	0.95
		A-TMLE	0.002253	0.018990	0.000365	0.95	0.96
	2500	AIPW	0.002303	0.074772	0.005585	0.96	0.95
		A-TMLE	0.000079	0.019475	0.000379	0.94	0.97
Scenario 2	1000	AIPW	0.004892	0.249615	0.062207	0.95	0.94
		A-TMLE	0.000092	0.030560	0.000932	0.95	0.97
	1500	AIPW	0.008641	0.143095	0.020510	0.96	0.94
		A-TMLE	0.000542	0.029893	0.000892	0.95	0.96
	2000	AIPW	0.000587	0.085222	0.007249	0.97	0.96
		A-TMLE	0.000773	0.019718	0.000389	0.96	0.97
	2500	AIPW	0.000958	0.072278	0.005215	0.95	0.94
		A-TMLE	0.000027	0.020994	0.000440	0.95	0.97
Scenario 3	1000	AIPW	0.007261	0.248291	0.061578	0.95	0.93
		A-TMLE	0.001035	0.033050	0.001091	0.94	0.96
	1500	AIPW	0.009303	0.143254	0.020567	0.95	0.94
		A-TMLE	0.001075	0.027192	0.000739	0.93	0.97
	2000	AIPW	0.000908	0.086972	0.007550	0.98	0.95
		A-TMLE	0.000396	0.022131	0.000489	0.95	0.97
	2500	AIPW	0.001882	0.069429	0.004814	0.96	0.94
		A-TMLE	0.000244	0.017762	0.000315	0.95	0.96

5 Cast study: LEADER trial subgroup analysis

To demonstrate the practical utility of our approach, we applied A-TMLE to estimate subgroup-specific treatment effects using data from the LEADER cardiovascular outcome trial (Marso et al., 2013, 2016). LEADER was a large, multinational, randomized, double-blind, placebo-controlled trial designed to evaluate the cardiovascular safety of liraglutide in patients with type 2 diabetes mellitus (T2DM) at high risk for cardiovascular events. The trial population consisted of T2DM patients 50 years of age or older with at least one pre-existing cardiovascular condition, with a glycated hemoglobin (HbA1c) level of 7.0% or greater, and who were either (1) treatment naive, or (2) treated with oral anti-hyperglycemic agents and/or insulin. the trial included $n = 9,340$ patients, $4,669$ in the liraglutide treatment group and $4,672$ in the placebo control group. Racial groups consisted of American Indian or Alaska Native ($n = 11$), Asian ($n = 936$), Black or African American ($n = 777$), Native Hawaiian or other Pacific Islander ($n = 8$), other ($n = 370$), or white ($n = 7238$). For the purposes of this subgroup analysis, we excluded groups with very small sample sizes ($n < 15$), as well as the (ill-defined) 'other' group. Mean exposure time in the trial was 3.5 years.

Study goal. Our goal was to estimate subgroup-specific risk differences in MACE between patients receiving the active intervention and those in the control arm across racial subgroups. Some subgroups are relatively underrepresented in the trial, leading to high variance and reduced statistical power for conventional subgroup-specific estimators. We apply A-TMLE to borrow strength from the remainder of the trial population while correcting for subgroup-specific differences, effectively treating observations outside each

subgroup as auxiliary data.

Estimator and its configurations. [MAYBE MOVE THIS TO APPENDIX] To use A-TMLE for estimating the target estimand, we used the `R` package `atmleFusion`. All nuisance components were estimated with the same discrete Super Learner library consisting of generalized linear models (Nelder and Wedderburn, 1972), Lasso regression (Tibshirani, 1996; Friedman et al., 2010), Bayesian additive regression trees (Chipman et al., 2010; Kapelner and Bleich, 2016), multivariate adaptive regression splines (Friedman, 1991), generalized additive models (Hastie and Tibshirani, 1986; Hastie, 2017), and random forests (Breiman, 2001; Wright and Ziegler, 2017), with the final learner selected by cross-validated log-likelihood. This library was used for the treatment mechanism $\bar{g}(W) = P(A = 1 \mid W)$, the outcome regression $\theta(W) = E(Y \mid W)$, the subgroup-membership $\Pi(A, W) = P(S = 1 \mid A, W)$, and the outcome regression $\bar{Q}(A, W) = E(Y \mid A, W)$. The working models for both the conditional average treatment effect $\tau_A(W) = E(Y \mid A = 1, W) - E(Y \mid A = 0, W)$ and the conditional average subgroup effect $\tau_S(A, W) = E(Y \mid S = 1, W, A) - E(Y \mid S = 0, W, A)$ were estimated using HAL with the group Lasso screening procedure described in Section 3.3, with first-order screening basis with `max_degree=5`, `smoothness_orders=1`, `num_knots=5`, followed by HAL fitting on the selected subspace with `max_degree=3`, `smoothness_orders=1`, and `num_knots=20`. The targeting method were implemented using the TMLE-style targeting described in Section XXX.

Results. We estimated the marginal average treatment effect comparing treatment arm to control arm on MACE events at 1, 1.5, 2, 3, and 4 years. We adjusted for the following baseline covariates: age, sex, diabetes duration, insurance(INSNVFL?), antihypertensive medication use (AHYPERFL), eGFR, BMI, HbA1c, LDL, HDL, and evidence of previous cardiovascular event (PRCVFL).

(to add descriptive statistics)

6 Discussion

This analysis demonstrates how A-TMLE enables valid, efficient estimation of subgroup-specific effects using only data already collected within the RCT, providing a scalable and interpretable tool for regulatory-relevant inference in trial populations with limited subgroup representation.

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A Canonical gradients of the parameters $\tilde{\Psi}$ and $\Psi^{\#}$

The target parameter considered in van der Laan et al. (2024a) captures the treatment effect in the RCT patients while averaging over the covariate distribution P_W of the pooled RCT and external RW patients. That is,

$$\Psi(P) = E_W E(Y_1 - Y_0 \mid S = 1, W).$$

Alternatively, one may be interested in generalizing the treatment effect to the RCT-only patients, that is,

$$\Psi_2(P) = E_W[E(Y_1 - Y_0 \mid S = 1, W) \mid S = 1],$$

but still would like to leverage external RWD to improve efficiency. For example, when one wants to use this A-TMLE to improve power of a subgroup analysis, one could define the subgroup of interest as $S = 1$, and any patients not in the subgroup as $S = 0$. In this case, one might actually want to average over the covariates of the subgroup rather than the pooled for two reasons. First, from an identification perspective, recall that if we go for parameter Ψ, then we need a positivity assumption $P(S = 1 \mid W) > 0$. However, one may run into settings where certain covariates either deterministically (or strongly) predict whether someone falls in the subgroup. Then there might be a violation (or practical violation) of this positivity assumption. Second, from an interpretation perspective, since our goal is to improve the analysis of this subgroup, it makes more sense to average over the subgroup we are interested in rather than average over the pooled population. Therefore, it is of interest here in this context to also develop an A-TMLE based estimator for the target parameter Ψ_2.

Similar to Lemma 2 in van der Laan et al. (2024a), we consider a decomposition of the target parameter. Specifically, we write $\Psi_2(P) = \tilde{\Psi}_2(P) - \Psi_2^{\#}(P)$, where

$$\tilde{\Psi}_2(P) = E_P[E_P(Y \mid A = 1, W) - E_P(Y \mid A = 0, W) \mid S = 1].$$

One could show that the bias parameter $\Psi_2^{\#}$ takes the form

$$\Psi_2^{\#}(P) = E_P[\Pi_P(0 \mid W, 0)\tau_{S,P}(W, 0) - \Pi_P(0 \mid W, 1)\tau_{S,P}(W, 1) \mid S = 1].$$

Note that in the special case when we only wish to augment with external controls, the bias parameter becomes

$$\Psi_2^{\#,\mathrm{EC}}(P) = E_P[\Pi_P(0 \mid W, 0)\tau_{S,P}(W, 0) \mid S = 1].$$

The following lemma presents the canonical gradient of the projection parameter $\tilde{\Psi}_{2,\mathcal{M}_{A,w}}$ at P.

Lemma A.1. *We define*

$$I_P = E_P[g_P(1-g_P)(1 \mid W)\phi_A\phi_A^\top(W)] \quad \textit{and} \quad \bar{Q}_{w,P}(W, A) = E_P(Y \mid W) + (A - g_P(1 \mid W)) \sum_j \beta_P(j)\phi_{A,j}(W).$$

The canonical gradient of $\tilde{\Psi}_{2,\mathcal{M}_{A,w}}$ at P is given by

$$D_{\tilde{\Psi}_2,\mathcal{M}_{A,w},P} = \frac{S}{P(S=1)}\left[\tau_{A,\beta_P}(W) - \tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P)\right] + \frac{1}{P(S=1)}\left\{E_P[S\phi_A(W)]\right\}^\top D_{A,\beta,P},$$

where

$$D_{A,\beta,P} = I_P^{-1}(A - g_P(1 \mid W))\phi_A(W)(Y - \bar{Q}_{w,P}(W, A)).$$

Proof. First, note that by the law of total expectation, we can express our target parameter $\tilde{\Psi}_{2,\mathcal{M}_{A,w}}$ as

$$\tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P) = \frac{1}{P(S=1)} E[I(S=1) \cdot \tau_{A,\beta_P}(W)].$$

We take the pathwise derivative of $\tilde{\Psi}_{2,\mathcal{M}_{A,w}}$ along a path P_ϵ through P at $\epsilon = 0$ with a corresponding score

$$s(O) = \frac{d}{d\epsilon} \log dP_\epsilon(O)\bigg|_{\epsilon=0}.$$

We have that

$$\frac{d}{d\epsilon}\tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P_\epsilon)\bigg|_{\epsilon=0} = \frac{d}{d\epsilon}\left[\frac{1}{P_\epsilon(S=1)} E_{P_\epsilon}\left[S \cdot \tau_{A,\beta_{P_\epsilon}}(W)\right]\right]_{\epsilon=0}.$$

Define

$$\mathbf{A}_\epsilon = P_\epsilon(S=1) \quad \text{and} \quad \mathbf{B}_\epsilon = \int S \cdot \tau_{A,\beta_{P_\epsilon}}(W) dP_\epsilon(O).$$

Then, we have

$$\begin{aligned}
\frac{d}{d\epsilon}\tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P_\epsilon)\bigg|_{\epsilon=0} &= -\frac{1}{\mathbf{A}^2} \cdot \frac{d}{d\epsilon}\mathbf{A}_\epsilon\bigg|_{\epsilon=0} \mathbf{B} + \frac{1}{\mathbf{A}} \cdot \frac{d}{d\epsilon}\mathbf{B}_\epsilon\bigg|_{\epsilon=0} \\
&= \frac{1}{\mathbf{A}}\left[\frac{d}{d\epsilon}\mathbf{B}_\epsilon - \tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P) \cdot \frac{d}{d\epsilon}\mathbf{A}_\epsilon\right]_{\epsilon=0}.
\end{aligned}$$

Recall the influence curve of $P(S=1)$ is $I(S=1) - P(S=1)$. Therefore, we have

$$\frac{d}{d\epsilon}\mathbf{A}_\epsilon\bigg|_{\epsilon=0} = E_P[I(S=1)s(O)].$$

Note also that

$$\frac{d}{d\epsilon}\tau_{A,\beta_{P_\epsilon}}(W)\bigg|_{\epsilon=0} = \frac{d}{d\epsilon}\left[\beta_{P_\epsilon}^\top \phi_A(W)\right]_{\epsilon=0} = \phi_A^\top(W)\frac{d}{d\epsilon}\beta_{P_\epsilon}\bigg|_{\epsilon=0} = \phi_A^\top(W) E_P[D_{A,\beta,P}(O)s(O)].$$

Therefore,

$$\begin{aligned}
\frac{d}{d\epsilon}\mathbf{B}_\epsilon\bigg|_{\epsilon=0} &= E_P\left\{S \cdot \tau_{A,\beta_P}(W)s(O) + S \cdot \phi_A^\top(W) E_P\left[D_{A,\beta,P}(O)s(O)\right]\right\} \\
&= E_P\left\{\left[S \cdot \tau_{A,\beta_P}(W) + E_P[S\phi_A(W)]^\top D_{A,\beta,P}(O)\right] s(O)\right\}.
\end{aligned}$$

Finally, we have that

$$\frac{d}{d\epsilon}\tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P_\epsilon)\bigg|_{\epsilon=0} = E_P\left\{\left[\frac{S}{P(S=1)}\left[\tau_{A,\beta_P}(W) - \tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P)\right] + \frac{1}{P(S=1)} E_P[S\phi_A(W)]^\top D_{A,\beta,P}(O)\right] \cdot s(O)\right\}.$$

Hence, the canonical gradient of $\tilde{\Psi}_{2,\mathcal{M}_{A,w}}$ at P is given by

$$D_{\tilde{\Psi}_{2,\mathcal{M}_{A,w},P}}(O) = \frac{S}{P(S=1)}\left[\tau_{A,\beta_P}(W) - \tilde{\Psi}_{2,\mathcal{M}_{A,w}}(P)\right] + \frac{1}{P(S=1)} E_P[S\phi_A(W)]^\top D_{A,\beta,P}(O).$$

□

The following lemma presents the canonical gradient of the bias projection parameter $\Psi^{\#}_{2,\mathcal{M}_{S,w}}$ at P.

Lemma A.2. *Let $\bar{\Pi}_P(W) = P(S = 1 \mid W)$. We define*

$$\tilde{I}_P = E_P \Pi_P(1 - \Pi_P)(1 \mid W, A)\phi_S \phi_S^\top(W, A) \quad \textit{and}$$
$$Q_{w,P}(S, W, A) = E_P(Y \mid W, A) + (S - \Pi_P(1 \mid W, A)) \sum_j \beta_P(j)\phi_{S,j}(W, A).$$

The canonical gradient of the bias projection parameter $\Psi^{\#}_{2,\mathcal{M}_{S,w}}$ at P is given by

$$D_{\Psi^{\#}_2,\mathcal{M}_{S,w},P}(O) = D_{\Psi^{\#}_2,\mathcal{M}_{S,w},P_W,P}(O) + D_{\Psi^{\#}_2,\mathcal{M}_{S,w},\Pi,P}(O) + D_{\Psi^{\#}_2,\mathcal{M}_{S,w},\beta,P}(O),$$

where

$$D_{\Psi^{\#}_2,\mathcal{M}_{S,w},P_W,P}(O) = \frac{S}{P(S=1)}\Big[\Pi_P(0 \mid W, 0)\tau_{S,\beta_P}(W, 0) - \Pi_P(0 \mid W, 1)\tau_{S,\beta_P}(W, 1)\Big] - \Psi^{\#}_{2,\mathcal{M}_{S,w}}(P),$$
$$D_{\Psi^{\#}_2,\mathcal{M}_{S,w},\Pi,P}(O) = \frac{\bar{\Pi}_P(W)}{P(S=1)}\left[\frac{A}{g_P(1 \mid W)}\tau_{S,\beta_P}(W, 1) - \frac{1-A}{g_P(0 \mid W)}\tau_{S,\beta_P}(W, 0)\right](S - \Pi_P(1 \mid W, A)),$$
$$D_{\Psi^{\#}_2,\mathcal{M}_{S,w},\beta,P}(O) = \frac{1}{P(S=1)}\Big[E_P[S \cdot \Pi_P(0 \mid W, 0)\phi_S(W, 0)] - E_P[S \cdot \Pi_P(0 \mid W, 1)\phi_S(W, 1)]\Big]^\top D_{S,\beta,P}(O).$$

where

$$D_{S,\beta,P}(O) = \tilde{I}_P^{-1}(S - \Pi_P(1 \mid W, A))\phi_S(W, A)(Y - Q_{w,P}(S, W, A)).$$

Proof. First, we again express our target parameter $\Psi^{\#}_{2,\mathcal{M}_{S,w}}$ as

$$\Psi^{\#}_{2,\mathcal{M}_{S,w}}(P) = \frac{1}{P(S=1)}E_P\Big\{S\left[\Pi_P(0 \mid W, 0)\tau_{S,\beta_P}(W, 0) - \Pi_P(0 \mid W, 1)\tau_{S,\beta_P}(W, 1)\right]\Big\}.$$

We define the control-arm portion and treatment-arm portion of the target parameter as

$$\Psi^{\#}_{2,\mathcal{M}_{S,w},A_0}(P) = \frac{1}{P(S=1)}E_P\Big\{S[\Pi_P(0 \mid W, 0)\tau_{S,\beta_P}(W, 0)]\Big\},$$
$$\Psi^{\#}_{2,\mathcal{M}_{S,w},A_1}(P) = \frac{1}{P(S=1)}E_P\Big\{S[\Pi_P(0 \mid W, 1)\tau_{S,\beta_P}(W, 1)]\Big\},$$

respectively, so that $\Psi^{\#}_{2,\mathcal{M}_{S,w}}(P) = \Psi^{\#}_{2,\mathcal{M}_{S,w},A_0}(P) - \Psi^{\#}_{2,\mathcal{M}_{S,w},A_1}(P)$. We focus on the control-arm portion as the treatment-arm portion is an exact mirroring of the same derivation. Similar to the previous proof, let

$$\mathbf{A}_\epsilon = P_\epsilon(S = 1) \quad \text{and} \quad \mathbf{C}_\epsilon = \int S \cdot [\Pi_{P_\epsilon}(0 \mid W, 0)\tau_{S,\beta_{P_\epsilon}}(W, 0)]dP_\epsilon(O).$$

Note that the influence curve of $\Pi_P(0 \mid w, a)$ is given by

$$\frac{I(W = w, A = a)}{P(w, a)}(I(S = 0) - \Pi_P(0 \mid w, a)).$$

So, we have

$$\frac{d}{d\epsilon}\Pi_{P_\epsilon}(0 \mid W,0)\bigg|_{\epsilon=0} = E_P\left\{\frac{1-A}{g_P(0 \mid W)}(I(S=0)-\Pi_P(0 \mid W,0))s(O)\bigg|W\right\}$$
$$= -E_P\left\{\frac{1-A}{g_P(0 \mid W)}(S-\Pi_P(1 \mid W,0))s(O)\bigg|W\right\}$$

As derived in Lemma 7 of our previous A-TMLE data fusion paper, the canonical gradient of β at P is given by

$$D_{S,\beta,P}(O) = \tilde{I}_P^{-1}(S-\Pi_P(1 \mid W,A))\phi_S(W,A)(Y-Q_{w,P}(S,W,A)).$$

So, we have

$$\frac{d}{d\epsilon}\tau_{S,\beta_{P_\epsilon}}(W,0)\bigg|_{\epsilon=0} = \phi_S^\top(W,0)E_P[D_{S,\beta,P}(O)s(O)].$$

Therefore, we have

$$\frac{d}{d\epsilon}\mathbf{C}_\epsilon\bigg|_{\epsilon=0} = E_P\left\{\left(S\cdot\Pi_P(0 \mid W,0)\tau_{S,\beta_P}(W,0) - E_P[S\cdot\Pi_P(0 \mid W,0)\tau_{S,\beta_P}(W,0)]\right)s(O)\right\}$$
$$- E_P\left\{\left(\bar{\Pi}_P(W)\tau_{S,\beta_P}(W,0)\frac{1-A}{\bar{g}_P(0 \mid W)}(S-\Pi_P(1 \mid W,0))\right)s(O)\right\}$$
$$+ E_P\left\{\left(E_P[S\cdot\Pi_P(0 \mid W,0)\Phi_S(W,0)]^\top D_{S,\beta,P}(O)\right)s(O)\right\}.$$

Recall from the previous proof that we have

$$\frac{d}{d\epsilon}\Psi^{\#}_{2,\mathcal{M}_{S,w},A_0}(P_\epsilon)\bigg|_{\epsilon=0} = \frac{1}{\mathbf{A}}\left[\frac{d}{d\epsilon}\mathbf{C}_\epsilon - \Psi^{\#}_{2,\mathcal{M}_{S,w},A_0}(P)\cdot\frac{d}{d\epsilon}\mathbf{A}_\epsilon\right]_{\epsilon=0}.$$

Hence, the canonical gradient of $\Psi^{\#}_{2,\mathcal{M}_{S,w},A_0}$ at P is

$$D_{\Psi^{\#}_2,\mathcal{M}_{S,w},A_0,P}(O) = \frac{S}{P(S=1)}\left[\Pi_P(0 \mid W,0)\tau_{S,\beta_P(W,0)} - \Psi^{\#}_{2,\mathcal{M}_{S,w},A_0}(P)\right]$$
$$- \frac{1}{P(S=1)}\bar{\Pi}_P(W)\tau_{S,\beta_P}(W,0)\frac{1-A}{g_P(0 \mid W)}(S-\Pi_P(1 \mid W,0))$$
$$+ \frac{1}{P(S=1)}E_P\left[S\cdot\Pi_P(0 \mid W,0)\phi_S(W,0)\right]^\top D_{S,\beta,P}(O).$$

The derivation for $D_{\Psi^{\#}_2,\mathcal{M}_{S,w},A_0,P}(O)$ mirrors this. We therefore arrive at the desired canonical gradient presented in the lemma. □

B TMLE-style targeting of the working model coefficients

In van der Laan et al. (2024b), the authors consider refitting the working-model coefficients so that the empirical mean of the corresponding components of the EIC is solved to zero. This is done by refitting the coefficients of the working models using the unpenalized version of the loss functions. In this appendix, we propose an alternative, TMLE-style approach for targeting these coefficients.

B.1 Targeting working model coefficients for the pooled-ATE parameter $\tilde{\Psi}$

Recall that the β-component of the EIC is given by

$$D_{\mathcal{M}_{A,w},\beta,P} = \tilde{D}_{\beta,P}^{\top} E_P \phi(W),$$

where

$$\tilde{D}_{\beta,P} = I_P^{-1}(A - g_P(1 \mid W))\phi(W) \left[Y - \theta_P(W) - (A - g_P(1 \mid W)) \sum_j \beta_P(j)\phi_j(W) \right],$$

and $I_P = E_P g_P(1 - g_P)(1 \mid W)\phi\phi^{\top}(W)$. Define $C_n = I_n^{-1} E_n \phi(W)$. Consider a one-dimensional submodel $\beta_{n,\epsilon} = \beta_n + \epsilon C_n$ for β through its initial β_n in the direction C_n. Then we have that

$$\tau_{A,\beta_{n,\epsilon}}(W) = \sum_j \beta_{n,\epsilon}(j)\phi_j(W) = \tau_{A,\beta_n}(W) + \epsilon\phi(W)C_n.$$

We define

$$H_n = (A - g_n(1 \mid W))\phi(W)C_n, \qquad R_n = Y - \theta_n(W) - (A - g_n(1 \mid W))\tau_{A,\beta_n}(W).$$

Then, we can represent the β-component of the EIC as $D_{\mathcal{M}_{A,w},\beta,P} = H_P R_P$. Note that

$$\begin{aligned} R_{n,\epsilon} &= Y - \theta_n(W) - (A - g_n(1 \mid W))\tau_{A,\beta_{n,\epsilon}}(W) \\ &= R_n - \epsilon(A - g_n(1 \mid W))\phi(W)C_n \\ &= R_n - \epsilon H_n. \end{aligned}$$

Therefore, the closed-form solution $\epsilon^{\star}$ of ϵ that solves the estimating equation $0 = P_n H_n(R_n - \epsilon H_n)$ is given by

$$\epsilon^{\star} = \frac{P_n(H_n R_n)}{P_n(H_n^2)}.$$

For $\tilde{\Psi}_2$, one could replace the definition of C_n by

$$I_n^{-1} E_n \left[\frac{S}{P_n(S = 1)} \phi_A(W) \right]^{\top}.$$

B.2 Targeting working model coefficients for the bias parameter $\Psi^{\#}$

Recall that the β-component of the EIC is given by

$$D^{\#}_{\mathcal{M}_{S,w},\beta,P} = D_{\beta,P}^{\top} \left[E_P \Pi_P(0 \mid W, 0)\phi(W, 0) - E_P \Pi_P(0 \mid W, 1)\phi(W, 1) \right],$$

where

$$D_{\beta,P} = I_P^{-1}(S - \Pi_P(1 \mid W, A))\phi(W, A) \left[Y - \theta_P(W, A) - (S - \Pi_P(1 \mid W, A)) \sum_j \beta_P(j)\phi_j(W, A) \right],$$

and $I_P = E_P \Pi_P(1 - \Pi_P)(1 \mid W, A)\phi\phi^\top(W, A)$. Define

$$C_n = I_n^{-1} \left[E_n \Pi_n(0 \mid W, 0)\phi(W, 0) - E_n \Pi_n(0 \mid W, 1)\phi(W, 1) \right].$$

Consider a one-dimensional submodel $\beta_{n,\epsilon} = \beta_n + \epsilon C_n$ for β in the direction C_n. Then we have that

$$\tau_{S,\beta_{n,\epsilon}}(W, A) = \sum_j \beta_{n,\epsilon}(j)\phi_j(W, A) = \tau_{S,\beta_n}(W, A) + \epsilon\phi(W, A)C_n.$$

Define

$$\begin{aligned} H_n &= (S - \Pi_n(1 \mid W, A))\phi(W, A)C_n \\ R_n &= Y - \theta_n(W, A) - (S - \Pi_n(1 \mid W, A))\tau_{S,\beta_n}(W, A), \end{aligned}$$

we can then write the EIC as $D^{\#}_{\mathcal{M}_{S,w},\beta,P} = H_P R_P$. Note that

$$\begin{aligned} R_{n,\epsilon} &= Y - \theta_n(W, A) - (S - \Pi_n(1 \mid W, A))\tau_{S,\beta_{n,\epsilon}}(W, A) \\ &= R_n - \epsilon(S - \Pi_n(1 \mid W, A))\phi(W, A)C_n \\ &= R_n - \epsilon H_n. \end{aligned}$$

Therefore, the closed-form solution $\epsilon^\star$ of ϵ that solves the estimating equation $0 = P_n H_n(R_n - \epsilon H_n)$ is again given by

$$\epsilon^\star = \frac{P_n(H_n R_n)}{P_n(H_n^2)}.$$

C Simulation study design

In this appendix, we describe details of our simulation study design, including the data-generating process (DGP), specifications of the candidate estimators, and definitions of the metrics used to evaluate the performance of estimators.

C.1 DGPs

We use the following method to simulate continuously distributed time-to-event data to mimic data from the real DEVOTE trial. Recall the proportional-hazard (PH) model with a Weibull baseline hazard, where the conditional event-time hazard is parameterized as

$$\lambda_\beta(t \mid A, L) = \eta\nu t^{\nu-1} \exp(f_\beta(A, L)),$$

where $\beta \to f_\beta(A, L)$. The corresponding cumulative hazard is

$$\Lambda_\beta(t \mid A, L) = \exp(f_\beta(A, L)) \int_0^t \eta\nu s^{\nu-1} ds = \eta t^\nu \exp(f_\beta(A, L)),$$

and the survival is then

$$S_\beta(t \mid A, L) = \exp(-\Lambda_\beta(t \mid A, L)) = \exp[-\eta t^\nu \exp(f_\beta(A, L))].$$

Set $S_\beta(T \mid A, L) = U$ and solve for T, we get

$$T = \left[\frac{-\log(U)}{\eta \exp(f_\beta(A, L))}\right]^{1/\nu}.$$

Therefore, to obtain a draw of the failure-time, we can draw from a uniform distribution $U \sim \text{Unif}(0, 1)$ and then apply the formula above.

To relax the PH assumption, we can make the function f time-dependent. For example, we could consider a piecewise PH model:

$$\lambda_\beta(t \mid A, L) = \begin{cases} \eta\nu t^{\nu-1} \exp(f_{\beta_1}(A, L)) & \text{if } 0 \le t < t_0 \\ \eta\nu t^{\nu-1} \exp(f_{\beta_2}(A, L)) & \text{if } t \ge t_0 \end{cases}$$

for $\beta_1 \neq \beta_2$ and some time point t_0. The cumulative hazard is then

$$\Lambda_\beta(t \mid A, L) = \begin{cases} \eta t^\nu \exp(f_{\beta_1}(A, L)) & \text{if } 0 \le t < t_0 \\ \eta[t_0^\nu \exp(f_{\beta_1}(A, L)) + (t^\nu - t_0^\nu) \exp(f_{\beta_2}(A, L))] & \text{if } t \ge t_0 \end{cases}$$

So, we could first compute $\Lambda_\beta(t_0 \mid A, L)$, if $-\log U < \Lambda_\beta(t_0 \mid A, L)$, then we can draw the failure-time from the first piece, otherwise we draw from the second piece.

DGP 1: simple proportional-hazard

$$
\begin{aligned}
L_1 &\sim \text{Unifnorm}(-1, 1) \\
(L_2, L_3) &\sim \text{Normal}\left(\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\right) \\
A &\sim \begin{cases} \text{Bernoulli}(\text{plogis}(5L_1)), & \text{scenario 1} \\ \text{Bernoulli}(\text{plogis}(L_1 + L_2 - 2L_3)), & \text{scenario 2} \\ \text{Bernoulli}(\Phi(0.5 + 0.25L_2 + 0.75L_3)), & \text{scenario 3} \end{cases} \\
\lambda(t \mid A, L) &= h_0(t) \exp(0.5A + 0.2L_1) \\
\lambda^c(t \mid A, L) &= 0.1
\end{aligned}
$$

DGP 2: complex proportional-hazard

$$
\begin{aligned}
L_1 &\sim \text{Uniform}(-1, 1) \\
(L_2, L_3) &\sim \text{Normal}\left(\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\right) \\
A &\sim \begin{cases} \text{Bernoulli}(\text{plogis}(5L_1)), & \text{scenario 1} \\ \text{Bernoulli}(\text{plogis}(L_1 + L_2 - 2L_3)), & \text{scenario 2} \\ \text{Bernoulli}(\Phi(0.5 + 0.25L_2 + 0.75L_3)), & \text{scenario 3} \end{cases} \\
\lambda(t \mid A, L) &= h_0(t) \exp(0.1A + 0.1AL_1^2 + 0.1 \sin(3\pi(L_2 + L_3)) + 0.1L_2 + 0.1L_3^3) \\
\lambda^c(t \mid A, L) &= h_0^c(t) \exp(0.1A + 0.05L_1 + 0.1L_2 + 0.04L_3)
\end{aligned}
$$

DGP 3: complex non-proportional-hazard

$$
\begin{aligned}
L_1 &\sim \text{Uniform}(-1, 1) \\
(L_2, L_3) &\sim \text{Normal}\left(\begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\right) \\
A &\sim \begin{cases} \text{Bernoulli}(\text{plogis}(5L_1)), & \text{scenario 1} \\ \text{Bernoulli}(\text{plogis}(L_1 + L_2 - 2L_3)), & \text{scenario 2} \\ \text{Bernoulli}(\Phi(0.5 + 0.25L_2 + 0.75L_3)), & \text{scenario 3} \end{cases} \\
\lambda(t \mid A, L) &= h_0(t) \exp(0.1A + 0.1AL_1^2 + I(t \geq 0.5)(0.1A + 0.1AL_1^2) + 0.1 \sin(3\pi(L_2 + L_3)) + 0.1L_2 + 0.1L_3^3) \\
\lambda^c(t \mid A, L) &= h_0^c(t) \exp(0.1A + 0.05L_1 + 0.1L_2 + 0.04L_3)
\end{aligned}
$$

D Implementation details